

SADEQ EBRAHIMI

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RESEARCH PROFILE

Electrical engineer with an M.S. in Communication Systems and research experience in statistical, sparse, array, and learning-based signal processing for wireless communications. Research contributions include published work on off-grid direction-of-arrival estimation in cluttered environments and a submitted manuscript on Faster-than-Nyquist (FTN) pulse shaping, with ongoing work on adaptive kernel-based methods for next-generation wireless communications and sensing systems.

RESEARCH INTERESTS

Primary Focus: Signal processing and learning for wireless communications and sensing.

Core Areas: Array and spatial signal processing; sparse and structured signal processing; learning-based and adaptive signal processing; advanced waveform and receiver design.

Current and Emerging Topics: Faster-than-Nyquist signaling, Predictive beamforming, integrated sensing and communications (ISAC), RIS/IRS-assisted systems, massive MIMO, channel estimation and prediction, adaptive kernel methods, information-theoretic learning, sparse recovery, and hardware-efficient physical-layer implementation.

EDUCATION

M.S. in Electrical Engineering – Communication Systems

Nov 2019

Sadjad University of Technology, Mashhad, Iran

GPA: 16.31 / 20

- **Thesis:** *Off-Grid DOA Estimation in the Presence of Clutter and Measurement Noise*
- **Advisor:** Dr. Ghazaleh Sarbishei

B.S. in Electrical Engineering – Electronics

Jan 2015

Imam Reza International University, Mashhad, Iran

GPA: 15.63 / 20

- **Thesis:** *Semiconductor Lasers*
- **Advisor:** Dr. Khosroabadi

PUBLICATIONS

Peer-Reviewed Journal Articles

- **Ebrahimi, S.**, Sarbishaei, G., & Hodtani, G. A. (2021). Off-grid direction of arrival estimation in the presence of measurement noise and heavy cluttered environment. *Signal, Image and Video Processing*, 15(4), 695–703. [doi:10.1007/s11760-020-01787-0](https://doi.org/10.1007/s11760-020-01787-0)

Manuscripts Submitted / Under Review

- **Ebrahimi, S.** “Cardinal Exponential-Spline Pulse Shaping for Faster-than-Nyquist Signaling.” Submitted to *IEEE Wireless Communications Letters*, 07 May 2026.

RESEARCH & ACADEMIC EXPERIENCE

Independent Research – Wireless Communications & Signal Processing

Ongoing

- Investigating adaptive kernel methods and lightweight learning techniques for predictive and data-driven physical-layer signal processing, with emphasis on computationally efficient approaches for wireless communications and sensing.

Graduate Researcher

2017 – 2019

Communication Systems Lab, Sadjad University of Technology

- Developed sparse off-grid DOA estimation methods for heavy-clutter and low-SNR environments, addressing grid mismatch and robustness limitations in conventional sparse recovery approaches.
- Designed numerical simulation frameworks to evaluate estimation accuracy, robustness, and computational complexity against benchmark sparse-recovery algorithms.

Undergraduate Research Assistant

Summer 2013

Electronics Laboratory, Imam Reza International University

- Assisted in testing and characterizing analog and digital electronic circuits using oscilloscopes, function generators, and spectrum analyzers.

TEACHING & MENTORING

Undergraduate Thesis Mentor

Summer 2018

Sadjad University of Technology

- Co-mentored undergraduate students on embedded DSP implementation using the Texas Instruments TMS320F2812 digital signal processor.

Teaching Assistant – Introduction to Communication Systems

Spring 2014

Imam Reza International University

- Held problem-solving sessions, assisted in grading coursework, and supported students with communication-systems concepts and exercises.

INDUSTRY EXPERIENCE

Embedded Systems Programmer

Summer 2017

Alia Co., Mashhad, Iran

- Programmed and debugged microcontroller-based embedded modules in C using Keil μ Vision.
- Collaborated on firmware verification and peripheral interfacing for industrial applications.

Electronics & Hardware Technician

Summer 2015

Independent, Mashhad, Iran

- Performed hardware troubleshooting, SMD soldering, and component-level diagnostic repairs on mobile communications devices.

TECHNICAL SKILLS

- **Programming & Simulation:** MATLAB (Signal Processing and Communications), Embedded C
- **Wireless & SDR Tools:** GNU Radio (basic flowgraph development and simulation)
- **Embedded:** ARM Cortex-M (STM32), ESP32, Arduino
- **FPGA & Hardware Acceleration:** Familiarity with Zynq SoC/FPGA architectures, high-level synthesis (HLS) workflows, and MATLAB/Simulink-based AMD Vitis Model Composer workflows

SELECTED ACADEMIC COURSEWORK

- **Graduate:** Wireless Communications (18.75/20), Cryptography (17.75/20), Channel Coding (17/20), Digital Image Processing (16.50/20)
- **Undergraduate:** Bachelor's Final Project (20/20), Signals & Systems (17.50/20), Digital Signal Processing (17/20)

LANGUAGES

- **English:** IELTS Academic – Overall Band 6.5 (August 2021)
- **Persian:** Native

REFERENCES

- **Prof. Ghosheh Abed Hodtani** – Professor, Department of Electrical Engineering, Ferdowsi University of Mashhad
Email: hodtani@um.ac.ir
- **Dr. Ghazaleh Sarbishei** – Assistant Professor, Sadjad University of Technology
Email: ghsarbishei@gmail.com
- **Dr. Somayeh Khosroazad** – Principal Systems & Design Engineer, NXP Semiconductors
Email: s.khosroazad@gmail.com